Air Ratchet



Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Air rachet

What it is used for

An air ratchet uses compressed air to tighten or loosen fasteners in automotive and workshop tasks. Use the correct socket, air pressure, and tool setting to work efficiently while preventing hand injuries, fastener damage, and flying debris.

Main parts

- Air ratchet body and drive square
- Forward/reverse selector
- Trigger or throttle control
- Air inlet and quick-connect fitting
- Air hose and coupling
- Compatible sockets and optional exhaust deflector

Before you start

1. Confirm that the air ratchet is suitable for the fastener and task.
2. Inspect the tool body, drive square, selector, trigger, air inlet, hose, and couplings for damage.
3. Check that the socket is the correct size, securely fitted, and rated for powered use.
4. Verify that the compressor and regulator supply clean, dry air within the tool manufacturer's pressure range.
5. Disconnect the air supply before fitting a socket, adjusting the tool, or performing inspection.
6. Wear safety glasses and hearing protection; secure loose clothing and tie back long hair.

How to use

1. Select the correct socket and push it fully onto the drive square.
2. Connect the tool to the air hose using the approved coupling.
3. Set the forward or reverse direction before operating the trigger.
4. Position the socket squarely on the fastener and keep fingers clear of pinch points.
5. Squeeze the trigger gradually and allow the ratchet to turn without forcing the tool.
6. Use the tool's rated capability and avoid applying excessive force to a stuck fastener.
7. For final tightening, use a calibrated torque wrench when the specified torque is critical.
8. Release the trigger fully before changing direction or moving to another fastener.
9. Disconnect the air supply before removing a jammed socket or inspecting the drive mechanism.
10. After use, disconnect the tool, clean external surfaces, and apply manufacturer-recommended air-tool lubricant if required.
11. Inspect the hose and fittings, then store the tool in a dry, secure location.

⚠ Safety

- Only trained users should operate pneumatic tools.
- Never exceed the tool's specified air pressure or use damaged hoses and couplings.
- Keep hands away from the ratchet head, rotating socket, and pinch points.
- Do not use cracked, worn, or unsuitable sockets.
- Disconnect the air supply before maintenance, socket changes, or clearing jams.
- Never direct exhaust air or flying debris toward people; use eye and hearing protection.

Care & storage

- Clean the tool after use and keep the air inlet free of debris.
- Lubricate with the specified air-tool oil only when required by the manufacturer.
- Inspect the drive square, selector, trigger, hose, and couplings regularly.
- Replace damaged hoses, worn sockets, and faulty fittings promptly.
- Arrange servicing if the tool loses power, leaks air, or operates irregularly.

Common mistakes to avoid

- Using excessive air pressure to increase tool power.
- Using the wrong socket size or a socket not rated for powered tools.
- Over-tightening fasteners without checking the specified torque.
- Changing sockets or clearing a jam while the air supply is connected.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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